

Response to reviewers

SMHL-D-26-00918, “Accuracy does not buy timing: a lead-time metric for stress anticipation in mobile and wearable sensing”

(revised title: “... in mobile sensing”; see Reviewer 2, comment 4)

Dear Editor and Reviewers,

Thank you for the careful reviews. They led us to re-examine the analysis pipeline end to end, not only the points raised. We summarise the main changes first, then answer each comment in turn. Section, table and figure numbers refer to the revised manuscript; in the tracked-changes version, additions are in blue and deletions in red.

Summary of main changes

1. **Stress trajectory.** ΔT is now computed on the expected stress level $s_t = \sum_c c p_t^{(c)}$, which represents stress directly, instead of the maximum class probability, which measures model certainty (R1-1, R2-2). The maximum-probability results move to Appendix A.
2. **One source for every number.** All StudentLife models were retrained, and every table and figure is generated by a single script from one set of records. This resolves the figure/table mismatches (R1-Q1, R1-Q2).
3. **New analyses:** null models (R2-1); onset-paired comparisons with a student-clustered bootstrap (R1-5); a TOST equivalence test with a stated margin (R1-3); a prediction-horizon experiment (R2-3); a full penalty $\times$ matching-window sensitivity grid (R1); exact p values and effect sizes with confidence intervals (R1).
4. **Narrower claims.** The equivalence claim is now limited to the quantity tested (the paired mean ΔT difference on onsets matched in both models). The chance-level statements follow the null-model results exactly. The forecast-trained models are compared with the present-state models on identical validation sequences. The paper also states that change points are located retrospectively, so ΔT measures temporal alignment and not real-time warning lead.
5. **Corrections we found ourselves.** (i) The reference implementation’s cross-validation is stratified by student and label at the example level, not split by subject; the text previously said the opposite. (ii) **ruptures** evaluates change points on a grid of every fifth sample by default; we now set `jump = 1`. (iii) The reference WESAD pipeline evaluates with a shuffled data loader and stores no window index, so the WESAD trajectories we analysed were not in temporal order. We have withdrawn the WESAD ΔT analysis and removed “wearable” from the title.

Reviewer 1

1. *The confidence trajectory does not consistently represent stress.*

Response. We agree. The maximum class probability is high whenever the model is confident, including confidently *unstressed*, so a change point in it need not be a change in predicted stress. We now use the expected stress level s_t (Eq. 2), which is monotone in the probability mass on higher stress classes. Appendix A reports the submitted max-probability trajectory and $P(\text{high})$ alongside s_t . Under the expected-stress trajectory the main result holds in the form we now state it: the paired mean ΔT differences between models are within the ± 6 h margin (Section 6.3), with anticipation

rates of 22.8–26.6% and medians of 0 h. Appendix A is a descriptive sensitivity table; we ran no equivalence test on the alternative trajectories. It shows the same pattern under each (medians of 0 h, anticipation rates of 22–27%).

Change. Section 5.1, “Stress trajectory”; Appendix A; terminology unified to “stress trajectory” throughout.

2. The asymmetric loss is not mathematically defined (temporal window, weights, normalisation, onset handling, role of λ).

Response. We now give the full definition. Onsets are upward label transitions in each training subject’s chronological sequence, computed from training labels only. Each training example receives weight $1 + \lambda$ if it lies within $H = 72$ h before its nearest onset, $(1 + \lambda)^{-1}$ if it lies within H after it, and 1 otherwise (Eq. 3). The weighted, class-weighted cross-entropy is averaged over the mini-batch without renormalisation (Eq. 4). The reconstruction term, validation loss and checkpoint selection are unchanged. $\lambda = 0$ recovers the reference objective; the pre/post weight ratio is $(1 + \lambda)^2$. The text also gives the numerical settings (class weights 0.6456, 0.5635, 1.0000; $\beta = 1$; $\alpha = 10^{-4}$ with an L1 reconstruction term summed over elements, which is absent in the decoder-free baseline used for this experiment; mini-batch size 1) and the tie rule (the earlier of two equally near onsets).

Change. Section 6.6, Eqs. 3–4.

3. The invariance claim needs an equivalence margin and an equivalence/non-inferiority test (e.g. TOST).

Response. Agreed: a non-significant difference is not evidence of equivalence. We pre-specify a margin of ± 6 h, a quarter of the daily EMA cycle and the smallest advance we consider actionable for a same-day intervention. We test equivalence with a TOST on the onset-paired mean difference, using a student-clustered bootstrap. Because a reader may prefer another margin, we also report the smallest margin at which equivalence holds for each pair. The margin was fixed when the revision analysis was designed, before the models were retrained. The TOST is computed on the student-level bootstrap distribution of the paired mean, so it accounts for several onsets coming from one student. For all three pairs the paired mean difference is within the ± 6 h margin (TOST $p \leq 0.005$), and in fact within 4.5 h (baseline vs. 1 branch), 3.3 h (baseline vs. 3 branches) and 2.8 h (1 branch vs. 3 branches). We have been careful about what this shows. It bounds the *mean* shift among onsets that both models respond to. It does not establish that the ΔT distributions have the same shape, tails or anticipation rate, and the abstract, contributions, Section 6.3 and conclusion now say “paired mean differences within the margin” instead of “equivalent distributions”.

Change. New Section 6.3, Table 4, Figure 5.

4. The cross-modal comparison is uncontrolled and relies on the anticipation rate, despite the paper’s own caution in Section 6.3.

Response. We agree, and have removed the cross-dataset claim. The abstract, contributions and conclusion no longer state a “24.8–33.3% band”. We went further than relabelling it as descriptive: for the reason given under Reviewer 2, comment 4, the WESAD ΔT analysis has been withdrawn altogether, so the revised paper makes no cross-modal claim and its title no longer refers to wearable sensing.

Change. Title; Abstract; Contributions; Results (WESAD subsection and the cross-modal table removed); Limitations (“Withdrawn cross-modal analysis”); Conclusion.

5. Observations are dependent; use paired comparisons of matched onsets across models.

Response. The folds depend only on the labels, so all configurations share identical validation sets

and identical onsets. For each onset we take the nearest change point of each model (median across repetitions) and compare models on the paired differences. Inference uses a bootstrap that resamples students, which handles dependence both within subjects and across repetitions. The paired mean differences are +2.16 h [95% CI -0.88 , $+5.10$] for baseline vs. 1 branch, $+0.43$ h [-3.03 , $+3.73$] for baseline vs. 3 branches, and -0.97 h [-3.14 , $+1.29$] for 1 vs. 3 branches (163–173 paired onsets from 22 students). The intervals and the TOST come from the student-level bootstrap. An onset-level Wilcoxon p value is also shown (0.51, 0.83, 0.09) but is labelled descriptive, because it treats onsets as independent. Onsets matched in one model but not the other are excluded from that pair, and the text says so. The pooled Wasserstein and Mann-Whitney results are kept as secondary descriptions and labelled as such.

Change. Section 6.3, Table 4.

6. Figures 2, 4 and 6 need formatting.

Response. All figures were regenerated from the new records, with consistent fonts, no overlapping titles, and vector output. The bar chart of Wasserstein distances (old Figure 4) is replaced by Figure 5, which shows the paired differences with confidence intervals against the equivalence margin. The asymmetric-loss figure (now Figure 7) was redrawn.

7. Proofreading: run-in headings in tables, repeated words, inconsistent terminology, spacing, wrong citations in the introduction.

Response. Corrected throughout. Specifically: run-in paragraph headings were reformatted; terminology is unified (“stress trajectory”, “anticipation rate”, “matched pair”, “onset”); the introduction now cites a paroxysmal-AF prediction study for cardiac lead time (Ebrahimzadeh et al., 2018) and a glucose-forecasting benchmark for prediction horizons (De Bois et al., 2022). Bibliographic errors were also fixed (Gjoreski et al. is 2017; Jackson et al. author list).

Minor: “Section ??” in the WESAD discussion.

Response. The unresolved reference was removed with the withdrawn WESAD subsection. Every remaining cross-reference was checked in the compiled PDF.

Q1: Figure 4 shows $W = 4.24/6.50/2.92$ h but Section 6.1 reports $3.55/5.75/2.64$ h. Q2: Figure 2 shows $24.5/25.5/30.2\%$ but Table 2 reports $24.8/25.6/30.7\%$.

Response. Thank you for catching these. In the submitted version the figures and the tables were produced by separate runs of the analysis, and the figure annotations pooled matched pairs across repetitions while the tables reported repetition means. We cannot reconstruct the submitted figures exactly, because the model outputs behind them are no longer available to us, so we did not try to patch the numbers. Instead we retrained every model and now generate every table and every figure from one set of records in a single pass, with figure annotations taken from the same values as the tables. The anticipation rate is reported both pooled and as a repetition mean $\pm$ SD, each labelled. The revised values also differ from the submitted ones for a reason we found during the revision and describe in Section 5.2: the submitted analysis used the **ruptures** default candidate grid of every fifth sample. Appendix A reports the submitted configuration re-run on the new models; it reproduces the scale of the submitted numbers (332–420 matched pairs, medians of $+4.0$ to $+5.5$ h, anticipation 24–25%).

Suggestions: exact p values and effect-size CIs instead of “ $p > 0.20$ ”; full asymmetric-loss maths; full StudentLife penalty and matching-window sensitivity.

Response. All three were done. Table 6 reports exact Mann-Whitney p (0.783 and 0.940) and, as the effect size, the onset-paired mean difference from the symmetric baseline with a 95% student-clustered

bootstrap CI (-0.13 h [$-2.73, +2.86$] and -1.78 h [$-5.26, +0.95$]). The Hodges-Lehmann shift is 0.0 h at both strengths. The macro-F1 gain is now 3.4 and 3.1 points (fold-paired Wilcoxon $p = 0.002$ and 0.010), with a small loss of micro-F1 that we report. The sensitivity grid (penalty $\in \{0.5, 1, 2, 5\} \times \text{window} \in \{24, 48, 72, 96\}$ h) is in Figure 9, and every cell, including penalty 5, is tabulated in Appendix B. Across penalty $\in \{0.5, 1, 2\}$ and window $\in \{24, 48, 72, 96\}$ h, the anticipation rate stays within 21–34% (21–28% at penalty ≤ 1), rises mechanically with the window, and the median ΔT is 0 h in every cell; no cell shows a personalisation advantage in timing. Penalty 5 leaves fewer than a dozen matched pairs, so those rates are not meaningful; they are listed in Appendix B for completeness.

Reviewer 2

1. (Major) *There is no null model: random change points or shuffled labels.*

Response. We added two nulls for each configuration, with 1,000 draws each. (a) *Random change points*: the same number of change points per subject, placed uniformly at random. (b) *Shifted labels*: each subject’s label sequence is circularly shifted, which preserves its autocorrelation and onset count while breaking alignment with the model. We compare the anticipation rate, the median ΔT and an alignment statistic (the share of change points within ± 24 h of an onset). No configuration shows significantly more anticipation than either null. Against random change points (mean $\approx 26\%$), the personalised models’ rates of 26.4% and 26.6% are within the null interval, and the baseline’s 22.8% is significantly *below* it ($p < 0.01$). Against shifted labels ($38\text{--}40\%$), every observed rate is far lower ($p < 0.01$). On alignment the picture is mixed, and we report it as such: the baseline’s change points fall within ± 24 h of an onset more often than random change points do (51.0% vs. 46.5% , $p < 0.01$) but not more often than under shifted labels ($p = 0.07$), and neither personalised model is above either null. The paper no longer says that a quarter of transitions carry a detectable precursor; a rate near 26% is what the matching rule gives without timing information.

Change. New Section 6.2, Table 3, Figure 4.

2. (Major) *Max-probability confidence tracks certainty, not stress; WESAD uses $P(\text{class } 1)$, which is inconsistent with StudentLife.*

Response. Agreed; see our response to R1-1. StudentLife now uses a stress trajectory, and the inconsistency disappears with the withdrawal of the WESAD analysis (comment 4).

3. *Models predict the current label, so the result is not surprising. Shift labels forward, retrain, and show how accuracy decays with horizon.*

Response. We accept the premise and ran the experiment. Labels were shifted 1, 2 and 3 EMA responses ahead (median gaps of 16, 30 and 49 h), and the baseline and three-branch models were retrained (5 repetitions $\times$ 5 folds each). ΔT is still measured against current-label onsets. Three things came out of it.

Accuracy does not decay with horizon. Micro-F1 goes from 0.478 to 0.490 for the baseline and from 0.611 to 0.626 for the three-branch model at the 49-hour gap. Self-reported stress is autocorrelated enough that forecasting is as easy as nowcasting.

A direct comparison with the $h = 0$ runs is misleading. Taken at face value, the mean ΔT is 3 to 6 h earlier and the anticipation rate 4 to 11 points higher in the forecast runs, and the difference in means is nominally significant in five of six runs. But the forecast runs are validated on different sequences (the last examples of each student are dropped and the folds change), and a random-change-point null on those sequences moves in the same direction with no model involved.

On identical sequences the difference disappears. Every example has an out-of-fold prediction from the present-state model, so we rebuilt the present-state model’s trajectory on each forecast run’s exact validation sequences and compared the two models onset by onset (Table 8). The paired mean differences are between -1.0 and $+2.0$ h with no consistent sign, every 95% interval includes zero, and the difference is within the ± 6 h margin in five of six runs (the baseline at $h = 2$ is the exception, with an interval reaching $+7.1$ h). Training the models to forecast up to 49 h ahead did not move their timing. We also report the lesson this taught us: ΔT summaries from different validation sequences are not comparable, and the Limitations now say so. We have also rephrased the framing: the paper now states explicitly that present-state models are not expected to anticipate, and that the contribution is a metric that makes this measurable and shows that the standard accuracy metrics do not reveal it.

Change. New Section 6.7, Tables 7 and 8, Figure 8; Limitations (“Dependence on the evaluation sequence”); Introduction, paragraph 4.

4. *WESAD/TSST: the 66 pairs share onsets and are not independent; median $\Delta T = 0$ is uninformative; the rate is 25.0% at $\rho = 1.0$, and “stable at 33%” cherry-picks three penalties; the abstract still uses 24.8–33.3%.*

Response. These criticisms are correct, and examining them uncovered a more basic problem. The reference WESAD pipeline evaluates with a shuffled data loader and stores no window index, so the trajectories we analysed were not in temporal order. The submitted WESAD ΔT values were therefore not valid timing measurements, and no reweighting of the 66 pairs could repair that. We have withdrawn the WESAD ΔT analysis. The paper now rests on StudentLife alone, the title no longer mentions wearable sensing, the “24.8–33.3%” band is gone from the abstract, contributions and conclusion, and the Limitations section states what was withdrawn and why. A cross-modal test with a temporally ordered evaluation is listed as the first item of future work.

Change. Title; Abstract; Sections 1, 3, 5 and 6; Limitations; Future directions; Conclusion.

5. *Writing: “Section ??”; overlapping title text in Figures 2 and 6; Islam et al. (2020, HARDC) is miscited as early arrhythmia warning.*

Response. All fixed; see R1-7 and R1-minor. HARDC is a heartbeat classification method, not an early-warning system, and has been replaced by Ebrahimzadeh et al. (2018) on predicting paroxysmal atrial fibrillation.

Further changes not requested by the reviewers

- **Anticipation-rate caution reworded.** The submitted text said that any widening of the ΔT distribution raises the anticipation rate. That is not true in general (it is unchanged for a distribution symmetric about zero), and Section 6.5 now says “can”. The section also no longer calls the rate difference between models a false positive: the difference in the rate is real; what it does not show is a shift in the median or the paired mean.
- **Retrospective change points.** Section 5 and the Limitations state that PELT is an offline procedure, so ΔT measures retrospective alignment and not the lead time of a real-time alert.
- **Macro-F1 mechanism.** The gain under the asymmetric loss is now reported with the accompanying micro-F1 loss, and class rebalancing is offered as an interpretation, since per-class scores were not recorded.

Sincerely,
David Olutunde Daniel